Predicted densities and weights in the historical CH instrument

1. Put the historical information on modern county units. Let P_c be historical population assigned to modern county c by geographic overlap. Let n_c and a_c be modern employment and land area in square kilometres. Counties combined in the employment data are kept together. Two historical density definitions are available: ℓ_c is either the population-weighted mean of historical source log densities within c, or $\log(P_c/a_c)$. The first preserves the source densities used by the original CH instrument. The second measures historical population on modern county land area.

2. Predict modern employment density from counties. For each historical year and territory scope, estimate by ordinary least squares

$$\log(n_c/a_c) = \alpha + \pi\ell_c + X_c'\gamma + v_c, \qquad \widehat{m}_c = \widehat{\alpha} + \widehat{\pi}\ell_c + X_c'\widehat{\gamma}.$$

Here X_c contains the selected geography and transport variables. The regression uses eligible counties across the contiguous United States, an intercept, and no fixed effects or human-capital terms. This is a prediction of employment density, separate from the county productivity regression used to adjust the modern CH index. The fitted value $\widehat{m}_c$ and predicted employment $\widehat{n}_c = a_c \exp(\widehat{m}_c)$ do not depend on θ.

3. Choose what the prediction replaces. For a modern state g, the three alternatives are

$$z_g^A = \frac{\sum_{c\in g} P_c \widehat{m}_c}{\sum_{c\in g} P_c}, \qquad z_g^B = \frac{\sum_{c\in g} \widehat{n}_c \ell_c}{\sum_{c\in g} \widehat{n}_c}, \qquad z_g^{A+B} = \frac{\sum_{c\in g} \widehat{n}_c \widehat{m}_c}{\sum_{c\in g} \widehat{n}_c}.$$

A changes the density, B changes the weights, and A+B changes both. Weights sum to one within each state. All three use the same county inputs. Their historical comparison on that county set is $\sum_c P_c \ell_c / \sum_c P_c$. The original historical option can retain additional counties when prediction inputs are missing.

4. Coverage and estimation. Prediction requires positive historical population, a defined density and complete selected geography. Estimating the county regression also requires positive modern employment. Coverage counts known historical area in the counties contributing to the index, relative to the state's full mapped area. The 80%, 95%, or no-cutoff choice applies in both historical years. It does not change the national county regression. Without a cutoff, small historical overlaps can enter; predicted weights use the full modern county land area.

The prediction uses modern employment, so these are estimated instruments. Reported standard errors and tests condition on the fitted prediction coefficients. They do not include uncertainty from estimating those coefficients or establish the exclusion restriction.